

SINGAPORE CHANGI INTL, Singapore

WMO: 486980

Lat: 1.3678N

Lon: 103.9826E

Elev: 7

StdP: 101.24

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.3	23.8	20.7	15.4	28.9	21.1	15.8	28.5	7.8	29.4	7.3	29.6	1.9	330	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.8	33.2	26.3	33.0	26.2	32.2	26.2	27.7	30.6	27.5	30.4	27.2	30.1	3.8	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.8	29.4	26.8	22.4	29.2	26.2	21.7	28.7	88.8	31.0	87.4	30.9	86.3	30.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
6.7	6.0	5.3	DB	22.0	34.4	0.5	0.7	21.7	34.9	21.4	35.3	21.2	35.7	20.8	36.3	
			WB	21.4	28.4	0.6	0.4	21.0	28.7	20.6	29.0	20.3	29.2	19.9	29.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	27.5	28.0	28.5	28.8	29.1	28.8	28.5	28.4	28.4	28.4	28.0	27.5
	DBStd	1.20	1.11	1.07	1.15	1.03	1.11	1.21	1.19	1.18	1.09	1.03	0.94	1.04
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6689	542	505	574	565	591	565	573	569	553	570	538	543
	CDD18.3	3647	283	272	316	315	333	315	315	311	303	312	288	284
	CDH23.3	39748	2649	2712	3372	3553	4001	3783	3674	3549	3445	3441	2896	2674
Wind	WSAvg	2.1	2.6	2.9	2.3	1.6	1.8	2.0	2.5	2.6	2.2	1.7	1.4	2.0
	PrecAvg	2054	222	131	159	146	161	136	142	144	137	162	249	299
	PrecMax	2887	601	418	446	308	371	266	305	327	440	425	536	766
	PrecMin	1119	15	0	23	25	42	22	19	45	24	11	73	63
	PrecStd	430	150	105	97	68	68	60	66	62	85	91	113	158
	DB	32.2	33.3	34.0	34.0	33.9	33.2	32.6	32.9	33.0	33.2	33.0	33.0	32.2
	MCWB	25.7	26.0	26.3	26.5	27.0	26.4	26.2	25.8	25.5	25.4	25.7	25.5	25.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	31.3	32.2	33.0	33.1	33.0	32.8	32.1	32.1	32.1	32.1	32.2	32.1	31.5
	MCWB	25.8	25.7	25.9	26.6	26.9	26.5	26.2	25.9	25.8	25.7	25.9	25.9	25.8
	DB	30.9	31.6	32.1	32.2	32.2	32.0	31.6	31.2	31.7	31.9	31.9	31.2	30.9
	MCWB	25.8	25.7	25.9	26.5	26.8	26.4	26.2	26.0	25.9	25.8	26.1	26.0	26.0
	DB	30.1	30.9	31.3	31.6	31.8	31.2	31.0	30.9	31.0	31.1	31.1	30.6	30.0
	MCWB	25.4	25.6	26.0	26.5	26.8	26.4	26.2	26.1	26.1	26.0	26.1	26.1	25.7
	WB	27.0	27.1	27.6	28.0	28.2	27.9	27.6	27.4	27.4	27.4	27.2	27.2	27.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	30.2	30.9	31.0	31.3	31.2	30.9	30.1	30.2	30.3	30.3	30.3	30.2	30.1
	WB	26.6	26.6	27.1	27.5	27.8	27.5	27.1	27.0	27.0	27.0	26.9	26.9	26.7
	MCDB	29.6	30.1	30.5	30.7	30.6	30.4	29.9	29.7	29.9	29.9	29.9	29.9	29.6
	WB	26.2	26.2	26.7	27.2	27.5	27.1	26.7	26.7	26.7	26.7	26.6	26.6	26.4
	MCDB	29.0	29.5	29.9	30.3	30.3	29.9	29.5	29.4	29.5	29.5	29.5	29.4	29.1
	WB	25.9	26.0	26.4	26.9	27.2	26.9	26.6	26.4	26.4	26.4	26.4	26.2	26.1
	MCDB	28.6	29.1	29.4	29.8	30.0	29.6	29.3	29.1	29.2	29.1	28.9	28.7	28.7
Mean Daily Temperature Range	MDBR	5.5	6.2	6.5	6.3	5.8	5.4	5.2	5.3	5.6	6.1	6.0	5.6	5.6
	5% DB	6.4	6.9	7.1	6.8	6.2	5.8	5.5	5.6	6.0	6.6	6.7	6.5	6.5
	MCWBR	2.3	2.2	2.3	2.5	2.5	2.4	2.3	2.3	2.4	2.5	2.6	2.6	2.6
	5% WB	5.8	6.3	6.5	6.4	5.9	5.5	5.3	5.4	5.6	6.1	6.2	5.9	5.9
	MCWBR	2.4	2.4	2.5	2.7	2.7	2.6	2.5	2.5	2.5	2.6	2.6	2.6	2.6
	taub	0.470	0.481	0.489	0.486	0.474	0.485	0.477	0.519	0.550	0.577	0.483	0.471	0.471
	taud	2.299	2.251	2.236	2.246	2.289	2.260	2.290	2.148	2.057	1.962	2.260	2.305	2.305
Clear-Sky Solar Irradiance	Ebn at Noon	861	857	845	829	817	794	807	790	783	766	843	854	854
	Edh at Noon	137	145	147	142	132	133	130	155	174	192	141	134	134
	RadAvg	4.55	5.27	5.26	5.09	4.71	4.45	4.43	4.50	4.50	4.49	4.23	4.13	4.13
All-Sky Solar Radiation	RadStd	0.38	0.47	0.46	0.32	0.23	0.32	0.35	0.26	0.30	0.34	0.27	0.31	0.31

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.37	+0.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page